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(54) **RETENTION ASSEMBLIES FOR MATERIAL STORAGE SYSTEMS, AND RELATED MATERIAL CONTAINERS AND METHODS**

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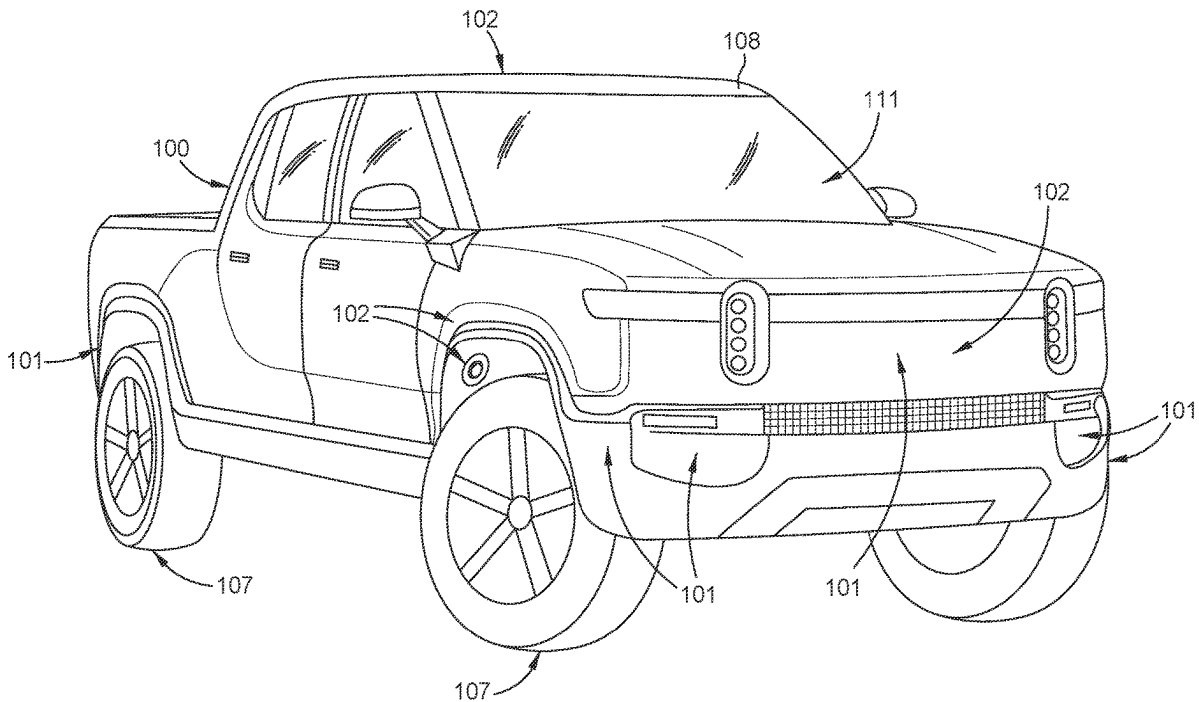
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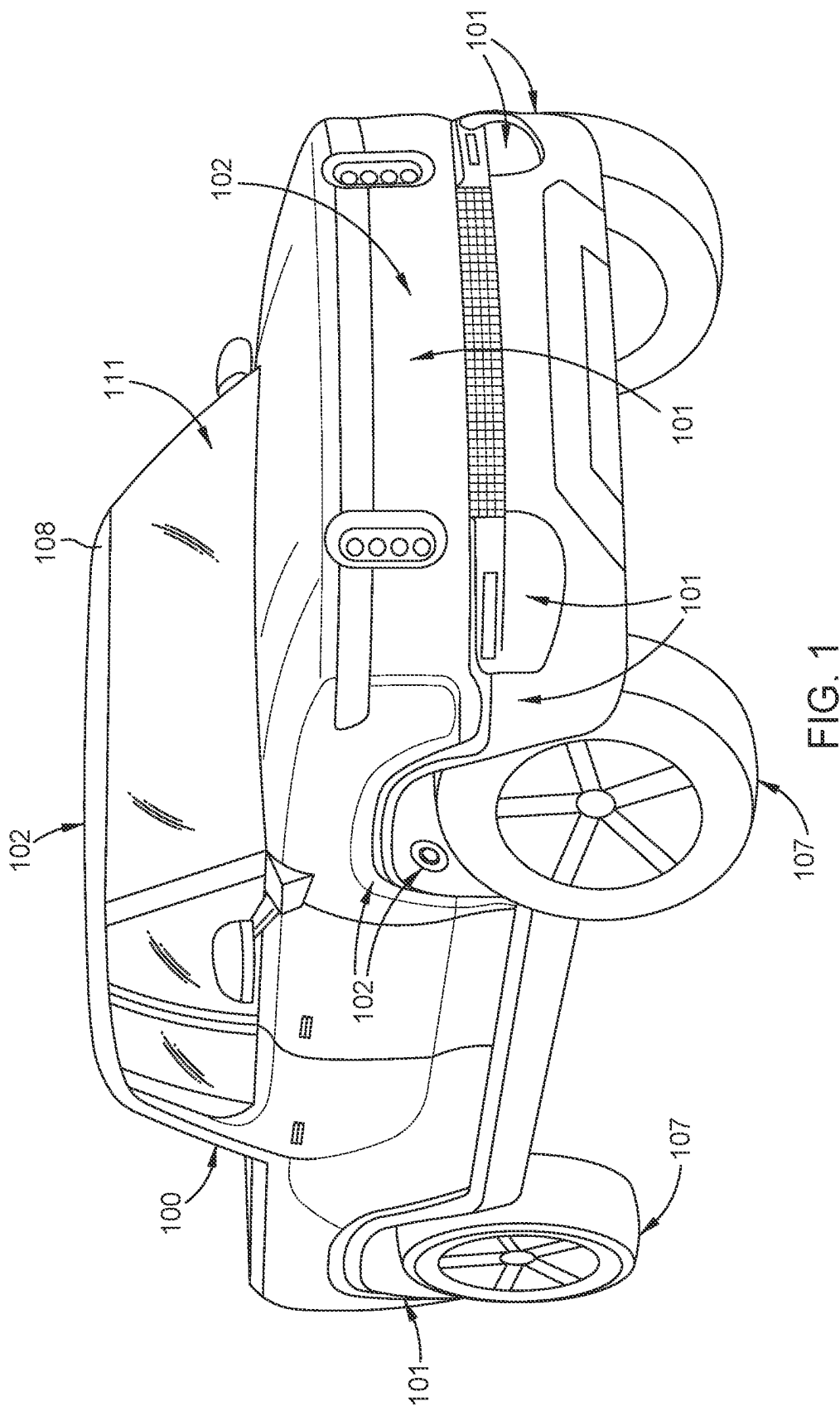
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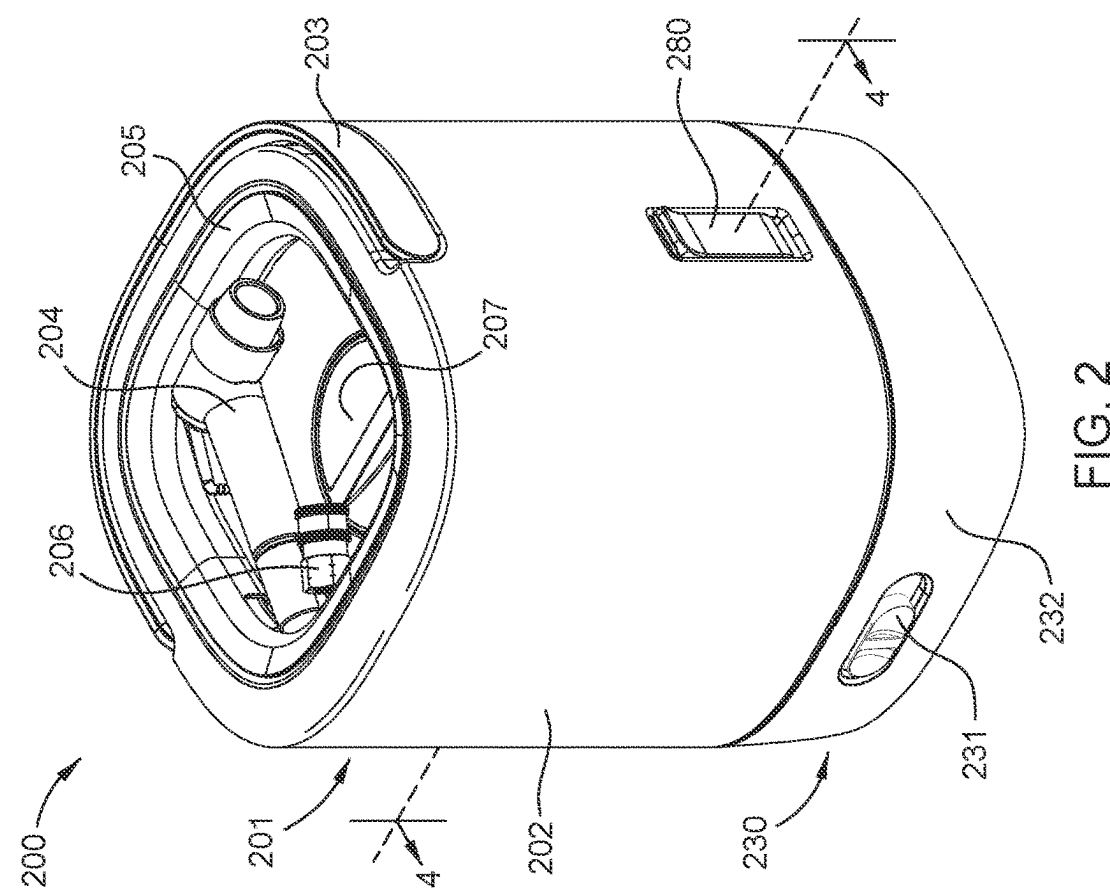
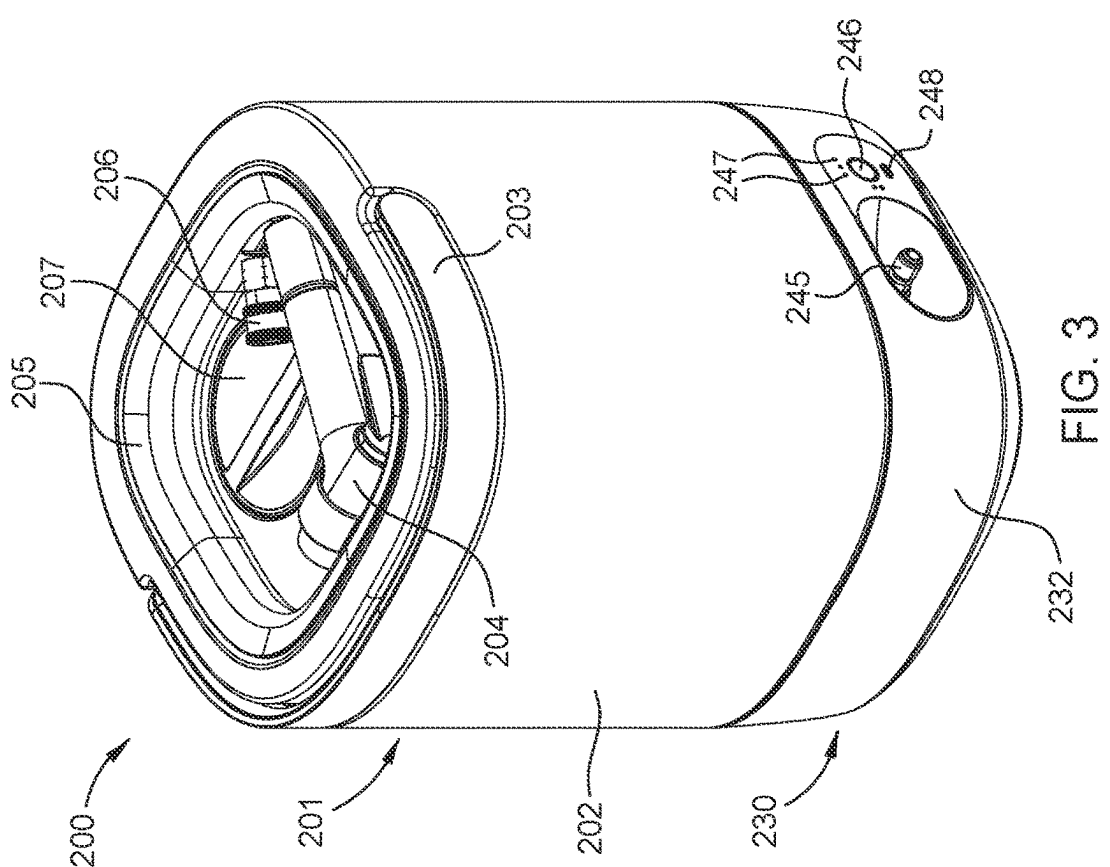
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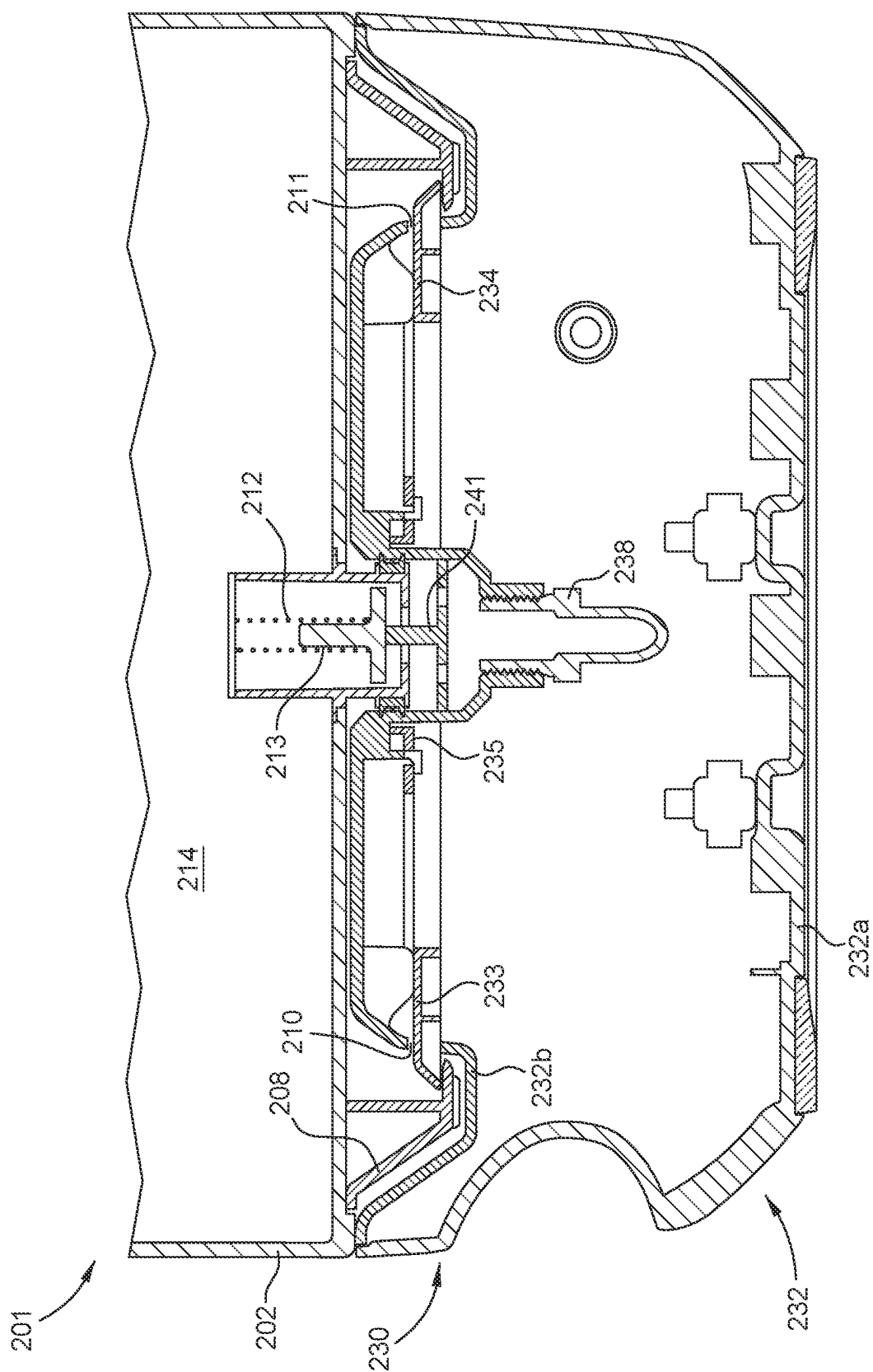
ABSTRACT

The present disclosure relates to retention assemblies for material storage systems, and related material containers and methods. In one or more embodiments, a retention assembly includes an actuator mounted to a retention housing, and a locking component coupled to the actuator. The actuator is movable to move the locking component between a locked position and an unlocked position. The retention assembly includes a magnet coupled to the actuator.









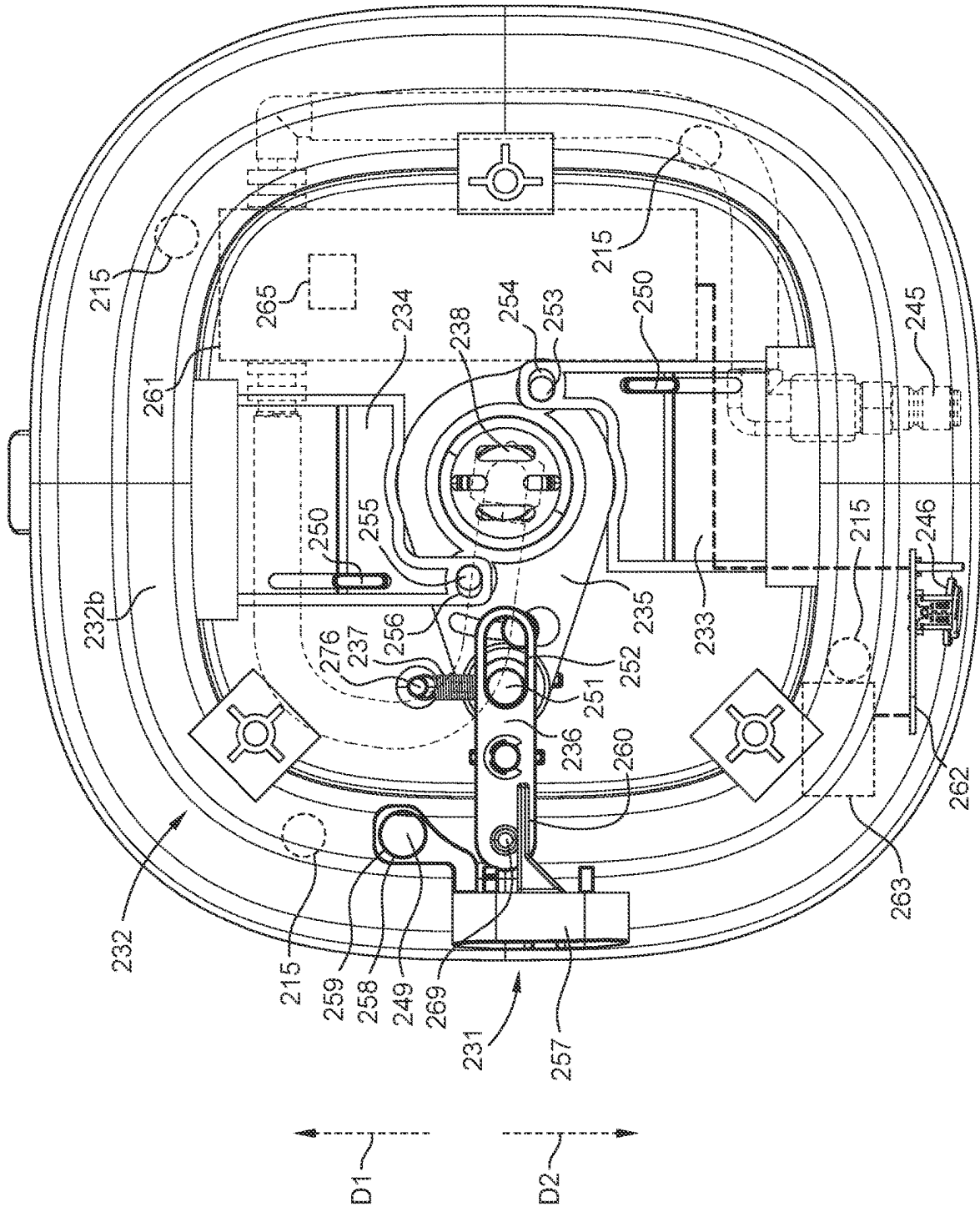
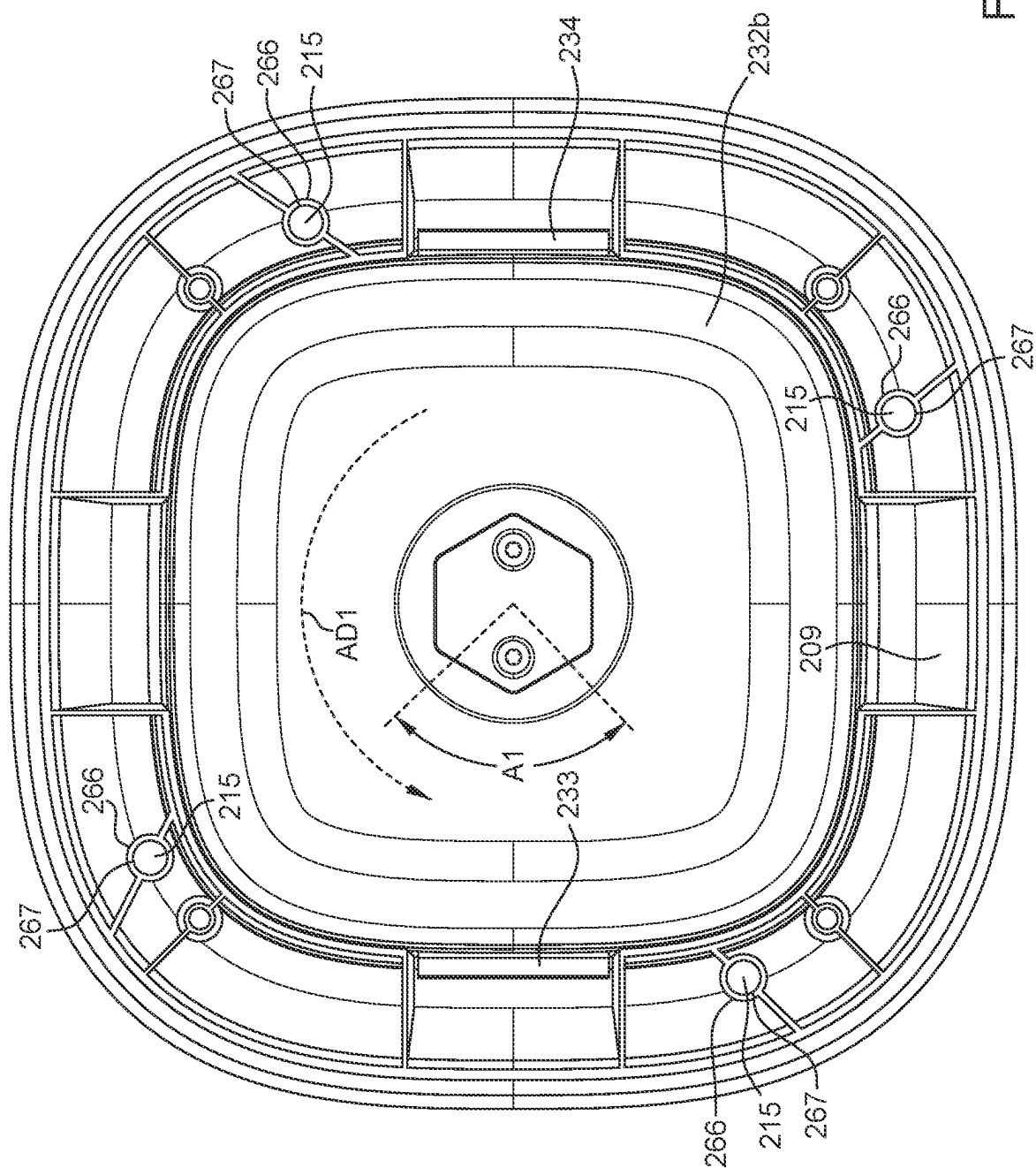


FIG. 5



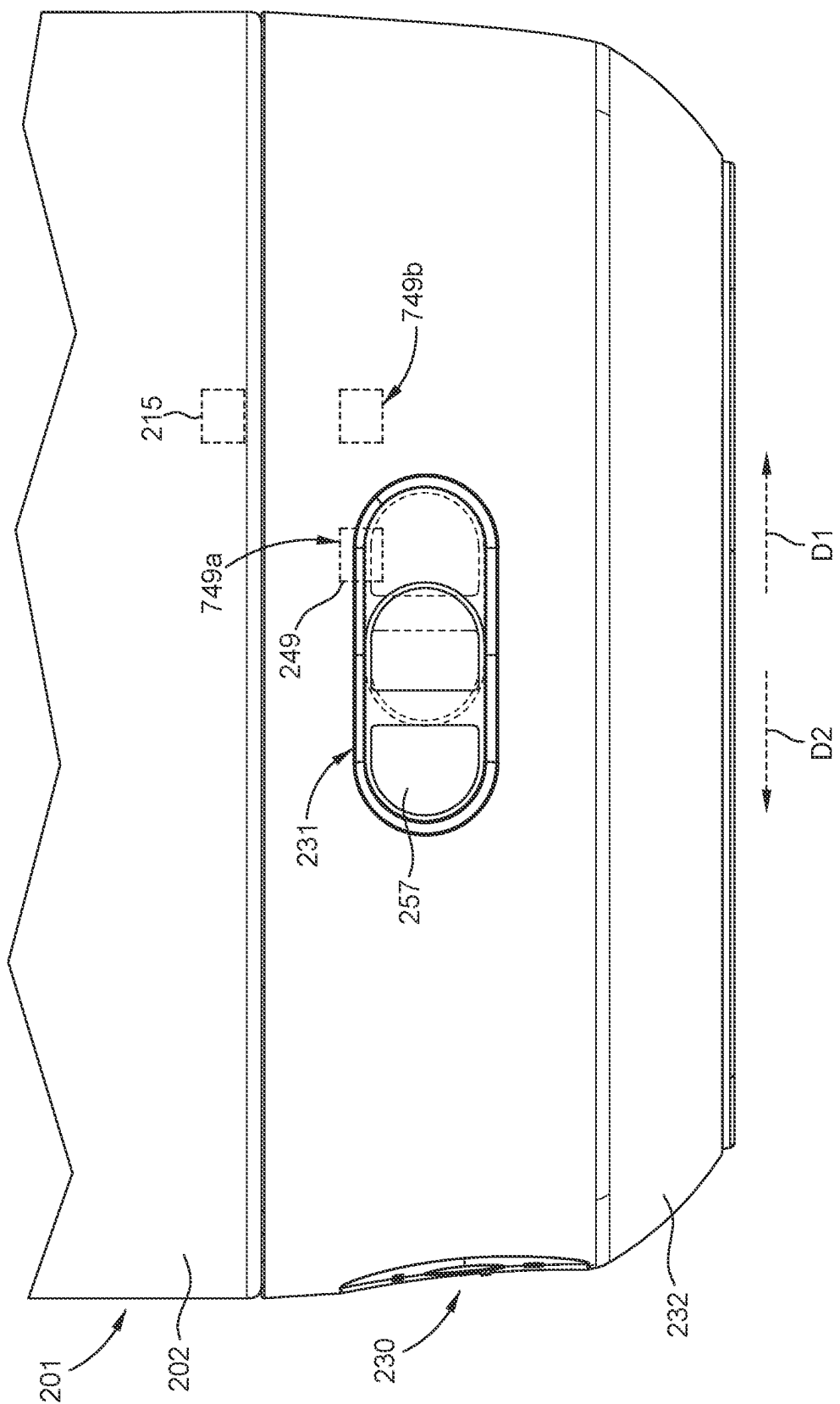
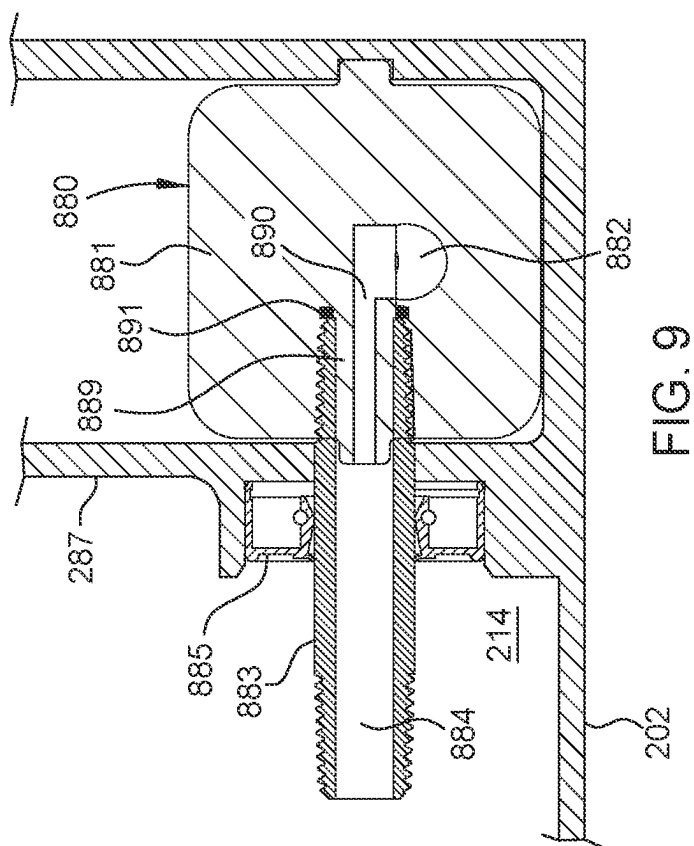
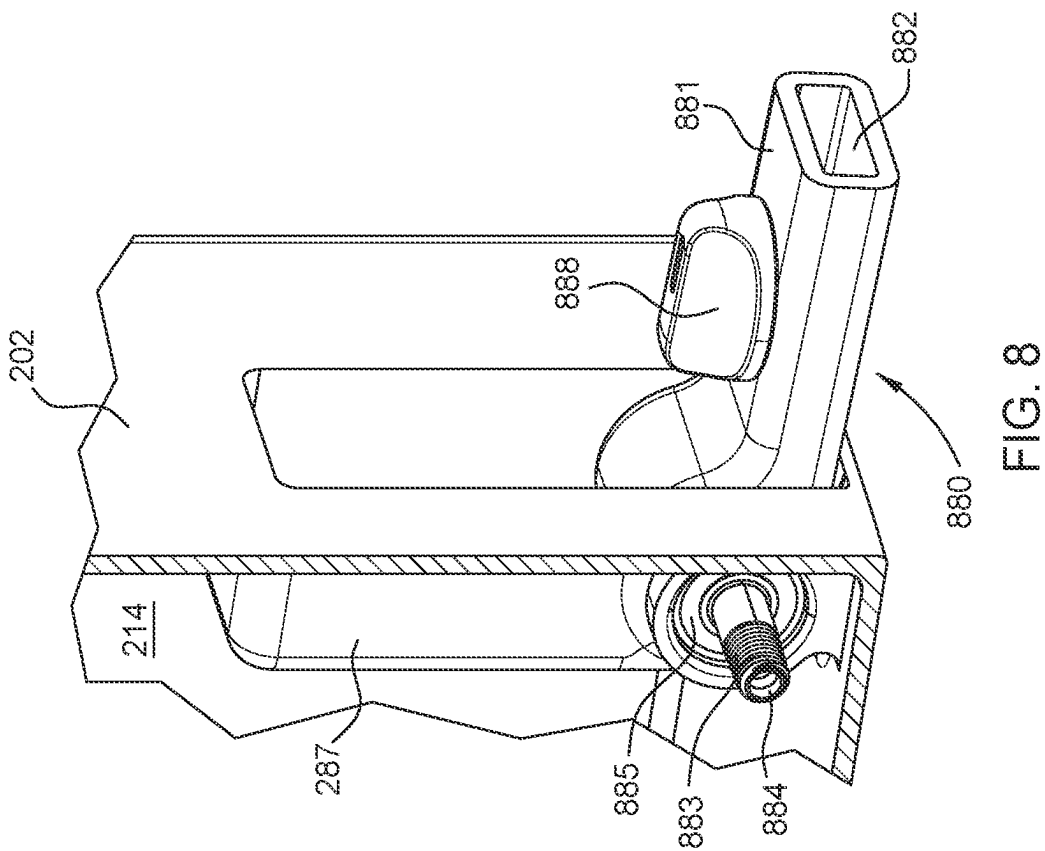


FIG. 7



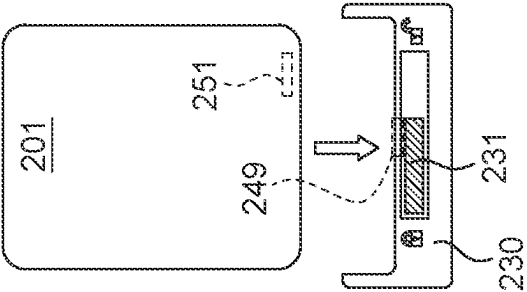


FIG. 10A

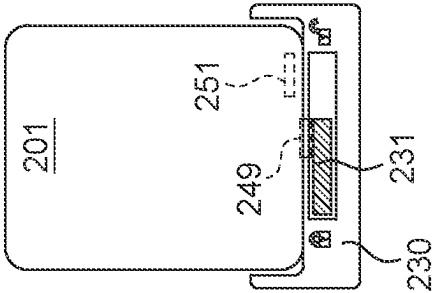


FIG. 10B

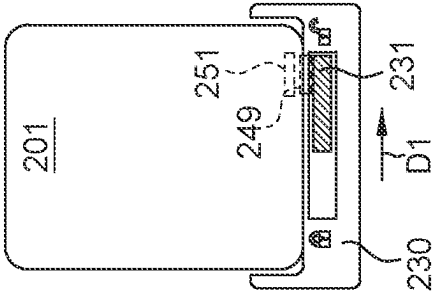


FIG. 10C

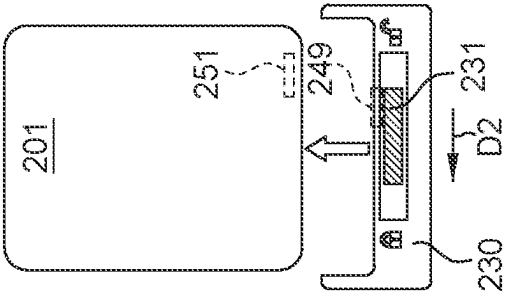


FIG. 10D

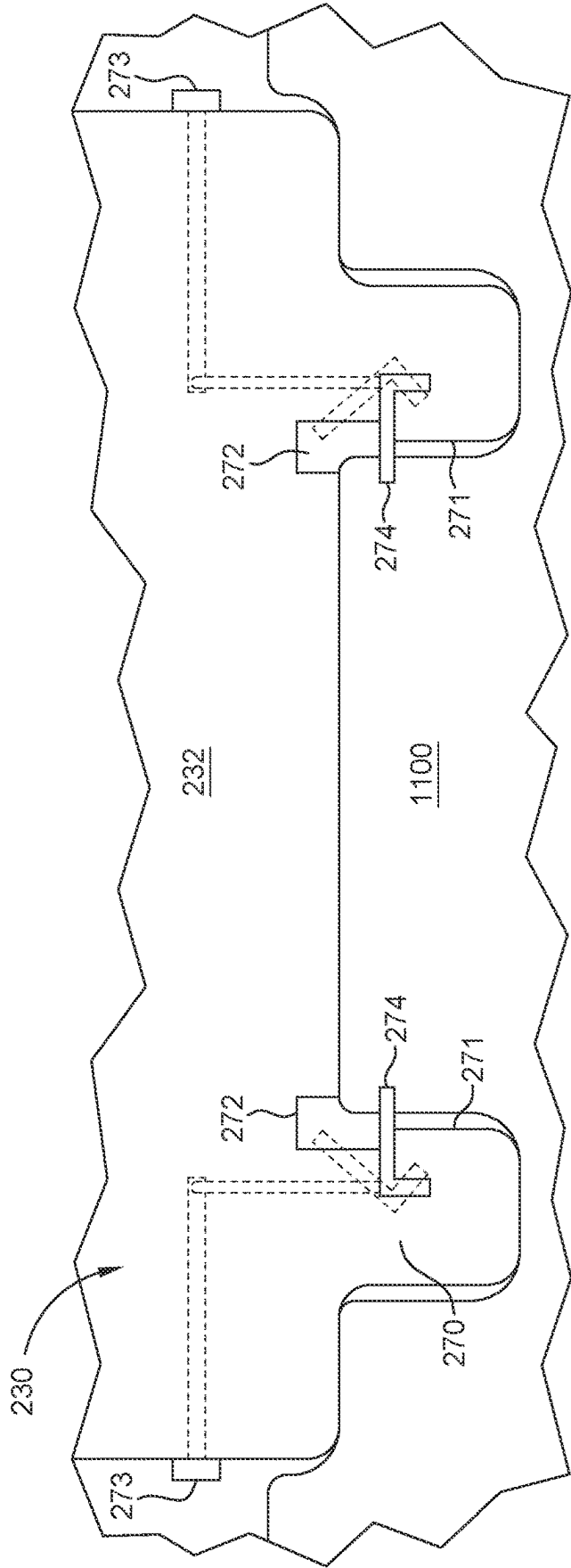


FIG. 11

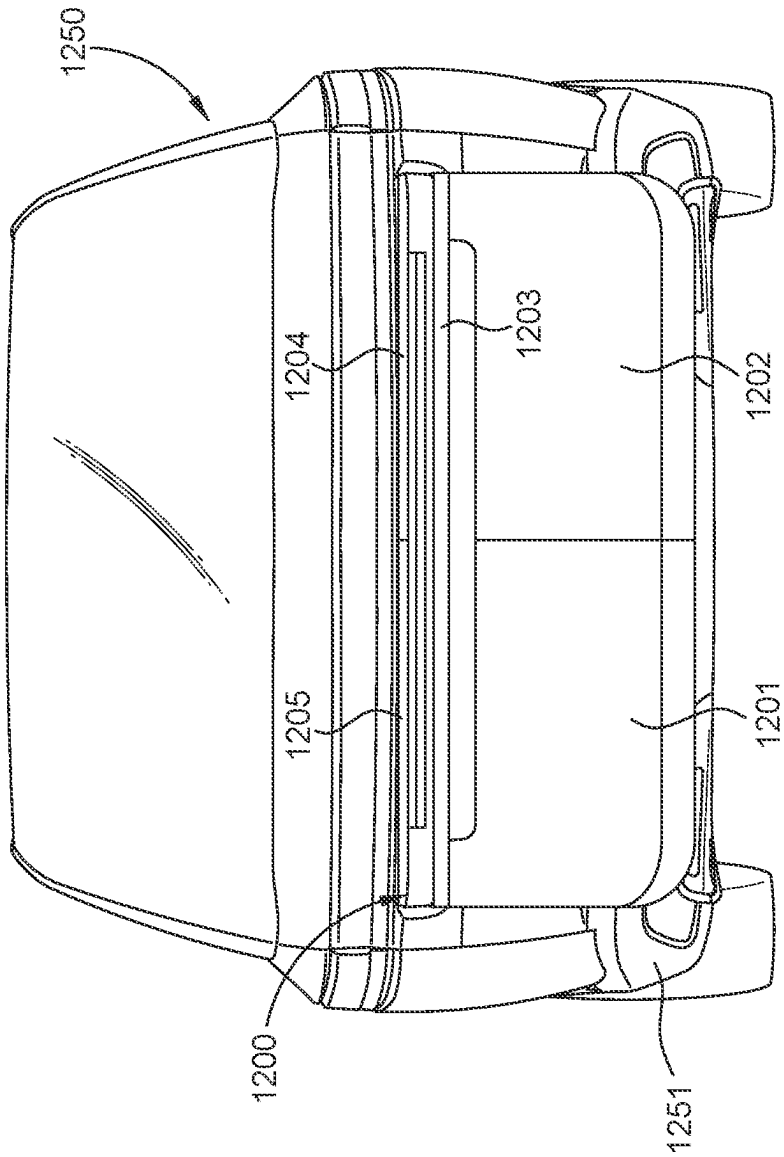
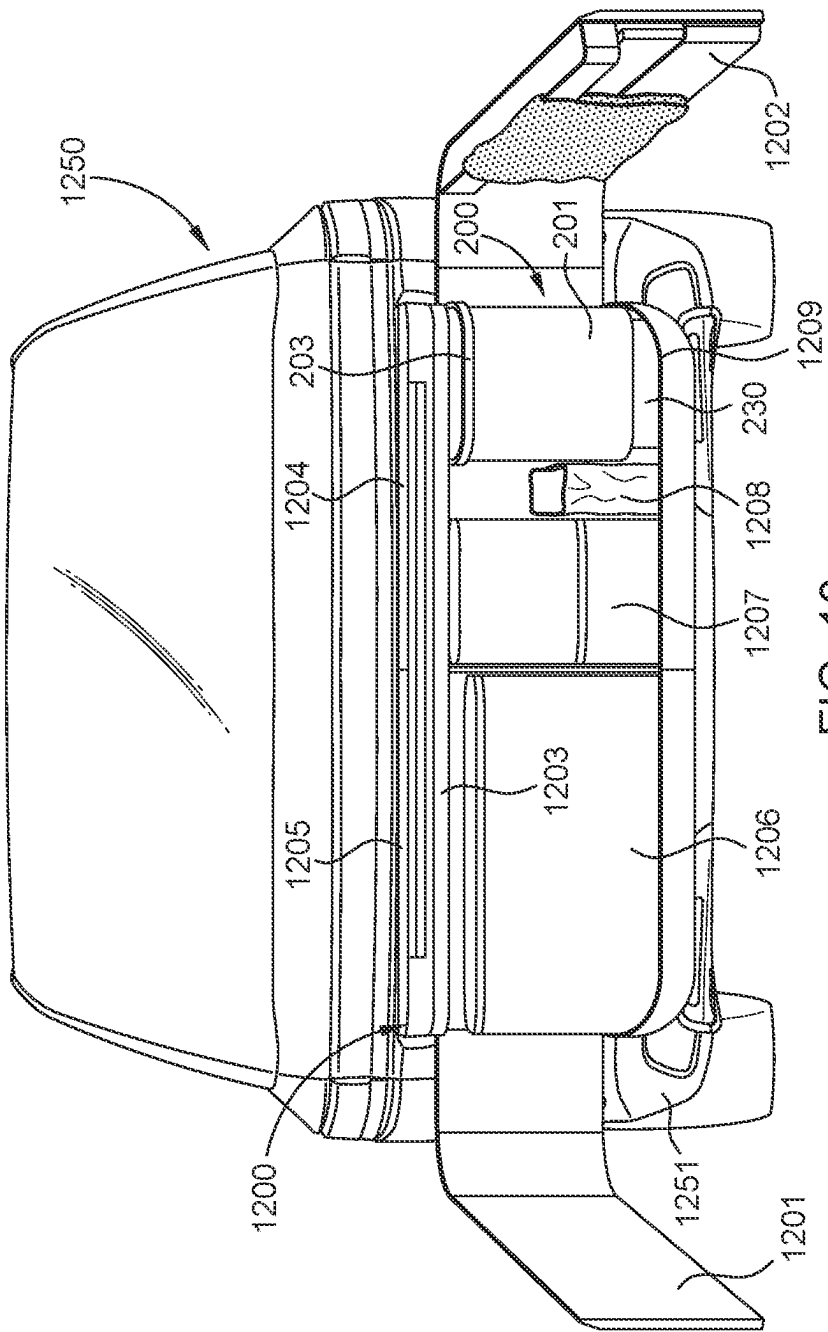


FIG. 12



RETENTION ASSEMBLIES FOR MATERIAL STORAGE SYSTEMS, AND RELATED MATERIAL CONTAINERS AND METHODS

INTRODUCTION

[0001] Material storage containers can be limited with respect to modularity and use. For example, it can be difficult and time consuming to refill storage containers because the storage containers can be heavy and large. Moreover, storage containers can be difficult to retain in place to prevent movement of the storage containers.

SUMMARY

[0002] The present disclosure relates to retention assemblies for material storage systems, and related material containers and methods.

[0003] In one or more embodiments, a retention assembly includes an actuator mounted to a retention housing, and a locking component coupled to the actuator. The actuator is movable to move the locking component between a locked position and an unlocked position. The retention assembly includes a magnet coupled to the actuator.

[0004] In one or more embodiments, a material container includes a storage volume defined at least partially by a storage housing, and a nozzle pivotably coupled to the storage housing. The nozzle includes a flow housing at least partially defining an outlet opening, and a shaft coupled to the flow housing and extending into a section of the storage housing. The shaft includes a flow opening between the storage volume and the outlet opening.

[0005] In one or more embodiments, a method of using a material storage system includes moving an actuator of a retention assembly to move a locking component of the retention assembly to an unlocked position to unlock a material container retained by the retention assembly. The moving of the actuator at least partially aligns a magnet coupled to the actuator with a second magnet coupled to the material container. The method includes lifting a material container off of the retention assembly, moving the material container away from a vehicle, and dispensing a material from the material container.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only exemplary embodiments and are therefore not to be considered limiting in scope, and may admit to other equally effective embodiments.

[0007] FIG. 1 is a schematic perspective view of a vehicle, according to one or more embodiments.

[0008] FIG. 2 is a schematic axonometric view of a first side of a material storage system, according to one or more embodiments.

[0009] FIG. 3 is a schematic axonometric view of a second side of the material storage system shown in FIG. 2, according to one or more embodiments.

[0010] FIG. 4 is a schematic partial cross-sectional view of the material storage system along Section 4-4 shown in FIG. 2, according to one or more embodiments.

[0011] FIG. 5 is a schematic partial bottom view of the retention assembly shown in FIGS. 2-4 with the base section hidden, according to one or more embodiments.

[0012] FIG. 6 is a schematic partial top view of the material container shown in FIGS. 2-5 with the storage housing hidden, according to one or more embodiments.

[0013] FIG. 7 is a schematic partial side view of the material storage system shown in FIGS. 1-6, according to one or more embodiments.

[0014] FIG. 8 is a schematic partial axonometric view of a nozzle, according to one or more embodiments.

[0015] FIG. 9 is a schematic cross-sectional view of the nozzle along Section 9-9 shown in FIG. 8, according to one or more embodiments.

[0016] FIGS. 10A-10D illustrate a schematic process flow of a method of using the material storage system, according to one or more embodiments.

[0017] FIG. 11 is a schematic partial side view of the retention assembly shown in FIGS. 2-10, according to one or more embodiments.

[0018] FIG. 12 is a schematic partial back view of a storage container coupled to a vehicle, according to one or more embodiments.

[0019] FIG. 13 is a schematic partial back view of the storage container shown in FIG. 12 with the one or more doors open, according to one or more embodiments.

[0020] To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements and features of one embodiment may be beneficially incorporated in other embodiments without further recitation.

DETAILED DESCRIPTION

[0021] The present disclosure relates to retention assemblies for material storage systems, and related material containers and methods.

[0022] In one or more embodiments, a material storage system includes a material container and a retention assembly. The material container includes a storage volume defined at least partially by a storage housing. The retention assembly is positionable to interface with the material container. The retention assembly includes an actuator mounted to a retention housing, and a locking component coupled to the actuator. The actuator is movable to move the locking component to lock and unlock the material container to and from the retention assembly. The retention assembly is operable to retain the material container when the material container is locked to the retention assembly. Additionally, the retention assembly includes a magnet coupled to the actuator, and the material container includes one or more second magnets.

[0023] The material storage system can be filled with fluid (such as liquid and/or gas), solids, and/or a combination (such as a slurry) thereof. The material storage system can store other materials. The material storage system can be connected to a spray nozzle. The fluid storage system can be used, for example, to spray off equipment and/or people (such as for cleaning and/or showering during camping), and/or to provide drinking water and/or cooking water. As described above, materials other than water may be in the material storage system. In one or more embodiments, a nozzle (e.g., a passive nozzle) is pivotably coupled to the storage housing. The nozzle includes a flow housing at least

partially defining an outlet opening, and a shaft coupled to the flow housing and extending into a section of the storage housing. The shaft includes a flow opening between the storage volume and the outlet opening.

[0024] In operation, the material container can be lowered onto the retention assembly such that the material container is locked to the retention assembly using the locking component. The material container can be unlocked from the retention assembly by moving the actuator to an unlocked position. In the unlocked position, the magnet coupled to the actuator aligns with at least one of the one or more second magnets included in the material container, such that a magnetic force (e.g., an attractive force) between the magnet and the second magnet(s) maintains the actuator in the unlocked position when the actuator is released. Hence, the material container can be lifted off of the retention assembly without needing to continuously hold the actuator in the unlocked position.

[0025] The material storage system and retention assembly enable quick and easy locking and locking of a material container to and from the retention assembly. For example, the actuator can be manipulated to the unlocked position and can remove the material container from the retention assembly after releasing the actuator. Moreover, the material container can be locked to the retention assembly by lowering the material container onto the retention assembly without manipulation of the actuator. The present disclosure contemplates that the operations described herein (such as the lowering, the lifting, the unlocking, and/or the manipulation) can be conducted manually by a user and/or automatically using automated equipment.

[0026] The disclosure contemplates that terms used herein such as “couples,” “coupling,” “couple,” and “coupled” may include but are not limited to bonding, embedding, welding, fusing, melting together, interference fitting, interference abutting, and/or fastening such as by using bolts, threaded connections, pins, and/or screws. The disclosure contemplates that terms such as “couples,” “coupling,” “couple,” and “coupled” may include but are not limited to integrally forming. The disclosure contemplates that terms such as “couples,” “coupling,” “couple,” and “coupled” may include but are not limited to direct coupling and/or indirect coupling, such as indirect coupling through components such as links, blocks, and/or frames.

[0027] FIG. 1 is a schematic perspective view of a vehicle 100, according to one or more embodiments. The vehicle 100 may include multiple sensors 101 and/or multiple cameras 102. The vehicle 100 includes one or more wheel modules 107. The vehicle 100 is an automotive vehicle. In the implementation shown in FIG. 1, the vehicle 100 is a truck, such as an electric truck. The present disclosure contemplates that the subject matter described herein can be used in any other type of vehicle having any numbers of wheels, such as vans, internal combustion vehicles, and/or sport-utility vehicles (SUVs). The vehicle 100 includes a view panel 111 mounted to a vehicle body 108 of the vehicle 100.

[0028] FIG. 2 is a schematic axonometric view of a first side of a material storage system 200, according to one or more embodiments. The material storage system 200 can be supported in or on a vehicle (such as the vehicle 100 shown in FIG. 1). The material storage system 200 can be supported in or on, for example, a bed, a cab, a hood, a trunk, a gear tunnel, a roof, and/or a floor of a vehicle. As another

example, the material storage system 200 can be supported in or on a storage container that is supported by and/or coupled to a vehicle.

[0029] The material storage system 200 includes a material container 201 and a retention assembly 230. The material container 201 includes a storage housing 202 and a carry handle 203 coupled to the storage housing 202. FIG. 2 shows the material container 201 as locked to the retention assembly 230. The material container 201 can be locked to and unlocked from the retention assembly 230. A spray nozzle 204 and a ring 205 are supported in a recess of the storage housing 202. A hose is connected to the spray nozzle 204 and a connector 206. The hose is wound about the ring 205. A nozzle 280 is pivotably coupled to the storage housing 202. The nozzle 280 is operable to flow a material in the material container 201. The nozzle 280 is shown as disposed above the nozzle 245 of the retention assembly 230. The nozzle 280 of the material container 201 can be moved to other positions (such as above the actuator 231). The material container 201 includes a removable cap 207 coupled to the storage housing 202. The removable cap 207 can be removed such that the material container 201 can be at least partially filled with the material.

[0030] The retention assembly 230 includes an actuator 231 mounted to a retention housing 232. FIG. 2 shows the actuator 231 in a locked position that locks the material container 201 to the retention assembly 230.

[0031] FIG. 3 is a schematic axonometric view of a second side of the material storage system 200 shown in FIG. 2, according to one or more embodiments.

[0032] The retention assembly 230 includes a nozzle 245 coupled to the retention housing 232, a power button 246, one or more electrical indicators 247, and an electrical port 248. The electrical port 248 can transmit data and/or electrical power to and from the retention assembly 230. As an example, the electrical port 248 can recharge the battery 263 described below. As another example, the electrical port 248 can transmit power from the battery 263 and to a mobile device (or other equipment) such that the retention assembly 230 can be used as a power bank. The electrical port 248 can be a USB-C port. Other ports are contemplated. The retention assembly 230 can be integrated with a vehicle (such as the vehicle 100) such that electrical power and/or data can be supplied from the power system of the vehicle and to the retention assembly 230 through the electrical port 248. The nozzle 245 is operable to flow the material in the material container 201. In one or more embodiments, the nozzle 245 of the retention assembly 230 is an active nozzle, and the nozzle 280 of the material container 201 is a passive nozzle.

[0033] FIG. 4 is a schematic partial cross-sectional view of the material storage system 200 along Section 4-4 shown in FIG. 2, according to one or more embodiments.

[0034] The retention assembly 230 includes a locking component 233 and a second locking component 234 coupled to the actuator 231. The actuator 231 is movable to move the locking component 233 and the second locking component 234 between the locked position and an unlocked position. The retention assembly 230 includes a rotatable link 235 pivotably coupled to the locking component 233 and the second locking component 234. In one or more embodiments, the retention housing 232 includes a base section 232a and a lid section 232b coupled together. The material container 201 includes an extension 208 extending relative to the storage housing 202, an opening

210 formed in the extension **208**, and a second opening **211** formed in the extension **208**. The actuator **231** is movable to move the locking component **233** into and out of the opening **210**. The actuator **231** is also movable to move the second locking component **234** into and out of the second opening **211**. FIG. 4 shows the locking components **233**, **234** in the locked position with the locking components **233**, **234** extending respectively into the openings **209**, **210**. The actuator **231** can be moved to the unlocked position such that the locking components **233**, **234** are retracted respectively out of the openings **209**, **210**. The storage housing **202** at least partially defines a storage volume **214** that is at least partially filled with the material.

[0035] The retention assembly **230** includes a column **241** disposed in a conduit **238**. The conduit **238** is fluidly connected to the nozzle **245** (shown in FIG. 3) of the retention assembly **230**. When the material container **201** is lowered onto the retention assembly **230**, the extension **208** pushes on angled (e.g., tapered) or curved outer surfaces of the locking components **233**, **234** to push the locking components **233**, **234** toward the rotatable link **235** until the locking components **233**, **234** can move into the openings **210**, **211** and into the locked position. Also, when the material container **201** is lowered, the column **241** pushes a plunger **212** against a bias element **213** to move the plunger **212** upward. The movement of the plunger **212** allows the material in the material container **201** to flow past the plunger **213**, through the conduit **238**, and out through the nozzle **245**.

[0036] In one or more embodiments, the actuator **231** includes a toggle button, the locking component **233** includes a first striker, the second locking component **234** includes a second striker, and/or the rotatable link **235** includes a pawl. FIG. 4 shows the nozzle **280** schematically and without hatching.

[0037] FIG. 5 is a schematic partial bottom view of the retention assembly **230** shown in FIGS. 2-4 with the base section **232a** hidden, according to one or more embodiments. The retention assembly **230** includes a magnet **249** coupled to the actuator **231**, and a transfer link **236** coupled between the rotatable link **235** and the actuator **231**. The rotatable link **235** is rotatable relative to the retention housing **232** and the storage housing **202**. The transfer link **236** is pivotable relative to the rotatable link **235** and the actuator **231** such that when the actuator **231** moves the transfer link **236** pivots to rotate the rotatable link **235**. The rotation of the rotatable link **235** moves the locking components **233**, **234**. In one or more embodiments, the locking components **233**, **234** are plates. The rotatable link **235** includes an extension **251** extending into an opening **252** of the transfer link **236**.

[0038] First extensions **250** of the lid section **232b** extend into channels formed in the locking components **233**, **234** to guide the locking components **233**, **234** as the locking components **233**, **234** extend toward the locked position and retract toward the unlocked position. The retention assembly **230** includes a bias element **237** operable to bias the actuator **231** toward the locked position. In one or more embodiments, the bias elements **213**, **237** respectively include a spring, such as a coil spring, for example a helical coil spring. The bias elements **213**, **237** can respectively be tape springs, wave springs, torsion springs, and/or flat springs. Other springs and/or other bias elements are contemplated. In one or more embodiments, the bias element **237** is under

compression and expands to bias the actuator **231** to the locked position. The bias element **237** is coupled to the extension **251** of the rotatable link **235** and a second extension **276** of the lid section **232b**. The magnet **249** is configured resist the bias element **237** to hold the actuator **231** in the unlocked position when an accessory (such as the material container **201**) is positioned on the retention assembly **230**.

[0039] The locking component **233** interfaces with a first side of the rotatable link **235**, and the second locking component **234** interfaces with a second side of the rotatable link **235**, the second side opposing the first side. The rotatable link **235** includes a second extension **253** extending into an opening **254** of the locking component **233**, and a third extension **255** extending into a second opening **256** of the second locking component **234**.

[0040] The actuator **231** includes a handle **257** and a receptacle **258** having an opening **259**. The magnet **249** is disposed in the opening **259**. In one or more embodiments, the magnet **249** is bonded (e.g., glued) to the actuator **231**. The actuator **231** is coupled to the transfer link **236** using a flange **260** abutting against a protrusion **269** of the transfer link **236**. The retention assembly **230** includes a pump **261** (shown in ghost for visual clarity purposes) fluidly connected to the nozzle **245**, a controller **262** in communication with the pump **261**, and a battery **263** (shown in ghost for visual clarity purposes) electrically connected to the controller **262**. The controller **262** can include, for example, a memory and/or a processor. The controller **262** can include a printed circuit board (PCB). The pump **261** is an electric pump **261**. The pump **261** pressurizes the material to pump the material through the nozzle **245** and the spray nozzle **204**. The material storage system **200** can include one or more heat transfer elements **265** operable to heat and/or cool the material. The one or more heat transfer elements **265** can, for example, be coupled to the retention assembly **230** (such as coupled to the pump **261**) to heat and/or cool the material as the material is dispensed and/or be coupled to the material container **201** to heat and/or cool the material as the material is stored in the material container **201**.

[0041] The material container **201** includes one or more second magnets **215** (a plurality, e.g., four, are shown). As the actuator **231** moves (e.g., along a first direction D1) to the locked position, the magnet **249** aligns under one of the second magnets **215** to generate and/or increase an attractive magnetic force between the magnet **249** and the respective second magnet **215**. In one or more embodiments, the magnet **249** and the one or more second magnets **215** are ferromagnetic. The attractive magnetic force is sufficient to overcome a bias force of the bias element **237** to maintain the actuator **231** in the locked position after the handle **257** is released by a user and/or automated equipment, which allows the material container **201** to be lifted off of the retention assembly **230** without continued manipulation of the handle **257**. During the lifting of the material container **201**, the respective second magnet **215** is moved away from the magnet **249** by a sufficient distance to reduce or eliminate the attractive magnetic force. When the attractive magnetic force is reduced or eliminated, the bias force of the bias element **237** moves the actuator **231** (e.g., along a second direction D2) back to the locked position shown in FIG. 5. The bias element **237** biases the locking component **233** toward the opening **210** (see FIG. 4) of the material

container **201** and biases the second locking component **234** toward the second opening **211** (see FIG. 4) of the material container **201**.

[0042] FIG. 6 is a schematic partial top view of the material container **201** shown in FIGS. 2-5 with the storage housing **202** hidden, according to one or more embodiments.

[0043] The one or more second magnets **215** are respectively disposed in opening(s) **267** of one or more protrusions **266** of the extension **208**. In one or more embodiments, the respective second magnets **215** are bonded (e.g., glued) to the one or more protrusions **266**. The second magnets **215** are spaced from each other along an azimuthal direction AD1. The second magnets **215** are spaced equidistantly from each other by an azimuthal angle A1 such that, if the material container **201** is rotated, the magnet **249** aligns with at least one of the one or more second magnets **215** when in the locked position. In one or more embodiments, the azimuthal angle A1 is about 90 degrees.

[0044] FIG. 7 is a schematic partial side view of the material storage system **200** shown in FIGS. 1-6, according to one or more embodiments.

[0045] FIG. 7 shows the actuator **231** in solid in the locked position, and in ghost in the unlocked position. The magnet **249** (shown in ghost) moves between a first position **749a** in the locked position and a second position **749b** in the unlocked position. In the locked position, the magnet **249** is offset from the respective second magnets **215**. In the unlocked position, the magnet **249** is at least partially aligned under at least one of the second magnets **215**.

[0046] FIG. 8 is a schematic partial axonometric view of a nozzle **880**, according to one or more embodiments. The nozzle **880** can be used in place of the nozzle **280** of the material container **201** shown in FIGS. 2 and 4.

[0047] FIG. 2 shows the nozzle **280** in a collapsed position, and FIG. 8 shows the nozzle **880** pivoted into an extended position. The nozzle **880** includes a flow housing **881** at least partially defining an outlet opening **882**. The nozzle **880** includes a shaft **883** coupled to the flow housing **881** and extending into a section **287** of the storage housing **202**. In one or more embodiments, the shaft **883** extends through the section **887** of the storage housing **202** and into the storage volume **214**. The shaft **883** includes a flow opening **884** between the storage volume **214** and the outlet opening **882**. A washer seal **885** is disposed about the shaft **883** of the nozzle **880**. The shaft **883** is rotatable relative to the storage housing. The nozzle **880** includes a valve **888** coupled to the flow housing **881**. The valve **888** can be, for example, an air return valve. Other types of valves are contemplated.

[0048] FIG. 9 is a schematic cross-sectional view of the nozzle **880** along Section 9-9 shown in FIG. 8, according to one or more embodiments.

[0049] In one or more embodiments, the shaft **883** is threaded into the flow housing **881** and is disposed about a rod section **889** of the flow housing **881**. An opening **890** of the rod section **889** fluidly connects the flow opening **884** of the shaft **883** to the outlet opening **882**. A second washer seal **891** is disposed about the rod section **289**. The rod section **889**, the shaft **883**, and the flow housing **881** can move (e.g., rotate) together. The rod section **889** can be integrally formed with the flow housing **881** or can be inserted into (e.g., threaded into) the flow housing **881**. The second washer seal **891** is disposed between the flow housing **881** and an end of the shaft **883**.

[0050] FIGS. 10A-10D illustrate a schematic process flow of a method of using the material storage system **200**, according to one or more embodiments.

[0051] The description of the method includes references to reference numerals of the material storage system **200**, and the present disclosure contemplates that subject matter (such as structures and components) other than the subject matter of the material storage system **200** can be used in relation to the method.

[0052] At FIG. 10A, the retention assembly **230** is shown with the actuator **231** in the locked position. The material container **201** is positioned in relation to (e.g., above) the retention assembly **230**.

[0053] At FIG. 10B, the material container **201** is positioned (e.g., lowered) to interface with the retention assembly **230** to lock the material container **201** to the retention assembly **230**.

[0054] At FIG. 10C, the method includes moving the actuator **231** to move the locking component **233** and the second locking component **234** to the unlocked position to unlock the material container **201** retained by the retention assembly **230**. The moving of the actuator **231** at least partially aligns the magnet **249** coupled to the actuator **231** with at least one of the second magnet **215** coupled to the material container **201**.

[0055] At FIG. 10D, the material container **201** is lifted off of the retention assembly **230**. The method can include moving the material container **201** away from a vehicle (such as the vehicle **100**). The lifting of the material container **201** moves the at least one second magnet **215** away from the magnet **249** to bias the actuator **231** to the locked position. FIG. 10D shows the actuator **231** as the actuator **231** is automatically biased toward the locked position.

[0056] The method can include dispensing the material from the material container **201**. For example, the material can be dispensed using the nozzle **280** and/or the nozzle **880** of the material container **201**. As another example, the material can be dispensed using the nozzle **245** of the retention assembly **230** when the material container **201** is interfacing with the retention assembly **230** (e.g., in FIG. 10B and/or FIG. 10C). The method can include connecting (e.g., using the connector **206**) the spray nozzle **204** to the nozzle **245** of the retention assembly **230** prior to the dispensing of the material.

[0057] In one or more embodiments, the method includes at least partially filling the storage volume **214** of the material container **201** with the material before FIG. 10A and/or after FIG. 10D when the material container **201** is lifted off of the retention assembly **230**. After the storage volume **214** is at least partially filled, the material container **201** can be lowered and/or re-lowered (as shown in FIG. 10B) onto the retention assembly **230** to lock the material container **201** to the retention assembly **230**.

[0058] When the fill level of the material container **201** is low, the material container **201** can be removed from the retention assembly **230** (as shown in FIG. 10D), and a second material container **201** (e.g., having a higher fill level) can be lowered and locked to the retention assembly **230** (as shown in FIG. 10B).

[0059] FIG. 11 is a schematic partial side view of the retention assembly **230** shown in FIGS. 2-10, according to one or more embodiments.

[0060] The retention assembly **230** is coupled to a vehicle panel **1100** of a vehicle. The retention assembly **230** includes

an extension 270 including an outer surface 271 and at least one opening 272 formed in the outer surface 271. The present disclosure contemplates that the extension 270 can be separated into a plurality of extensions. The retention assembly 230 includes at least one second actuator 273 mounted to the retention housing 232, and at least one third locking component 274 disposed in the at least one opening 272 and coupled to the at least one second actuator 273.

[0061] The at least one second actuator 273 is movable to move the at least one third locking component 274 between a first position (shown in solid in FIG. 11) and a second position (shown in ghost in FIG. 11). In the first position, the at least one third locking component 274 is at least partially outside of the at least one opening 272 and abutting against the vehicle panel 1100. In the second position the at least one third locking component 274 is moved into the at least one opening 272 relative to the first position. The implementation in FIG. 11 shows two third locking components 274 and two second actuators 273. The respective second actuators 273 are rotatable to pivot the respective third locking components 274 between the first position and the second position. In the first position, the third locking components 274 extend into one or more openings of the vehicle panel 1100 to lock the retention assembly 230 to the vehicle panel 1100. The second actuators 273 can include cams having key slots such that a key can be used to turn the second actuators 273 and lock the third locking components 274 in the first position or the second position. The present disclosure contemplates that the second actuators 273 can include pivotable bars. The vehicle panel 1100 can be disposed along a bed, a cab, a hood, a trunk, a gear tunnel, a roof, and/or a floor of a vehicle. As another example, the vehicle panel 1100 can be disposed along a storage container that is supported by and/or coupled to a vehicle.

[0062] The retention assembly 230 can be movable relative to a vehicle (as shown in FIGS. 2-10) or locked in place relative to the vehicle (as shown in FIG. 11). The retention housing 232 can be integrally formed with the vehicle panel 1100.

[0063] FIG. 12 is a schematic partial back view of a storage container 1200 coupled to a vehicle 1250, according to one or more embodiments. The storage container 1200 can be attached, for example, to a rear bumper 1251 of the vehicle 1250. The vehicle 1250 is shown as an SUV. Other vehicles are contemplated.

[0064] The storage container 1200 includes one or more doors 1201, 1202 (two are shown) a roof 1203, and a bench 1204. The bench 1204 can include a cook top 1205 (such as a stove top). The bench 1204 can be omitted and/or the roof 1203 can be moved to provide access inside of the storage container 1200. The one or more doors 1201, 1202 can be opened e.g., pivoted) to provide access inside of the storage container 1200. The present disclosure contemplates that the one or more doors 1201, 1202 can be replaced with stationary wall(s). One or more support bars can be coupled to the storage container 1200 and the vehicle 1250 such that the vehicle 1250 supports the storage container 1200. The one or more support bars can extend at least partially between the vehicle 1250 to space the storage container 1200 from the vehicle 1250.

[0065] FIG. 13 is a schematic partial back view of the storage container 1200 shown in FIG. 12 with the one or more doors 1201, 1202 open, according to one or more embodiments.

[0066] A cooler 1206, a trash container 1207, an equipment container 1208 (containing, e.g., pots and pans and/or kitchen utensils), and/or the material storage system 200 can be stored in the storage container 1200. The doors 1201 can also hold equipment (such as cooking ingredients, kitchen utensils, and/or other equipment). The retention assembly 230 is coupled to a floor 1209 of the storage container 1200. The retention housing 232 can be integrally formed floor 1209. The vehicle panel 1100 described in relation to FIG. 11 can be part of the floor 1209, and the retention assembly 230 can be removably locked to the floor 1209 in a manner—for example—described in relation to FIG. 11. Although not shown in FIG. 13, the present disclosure contemplates that the actuator 231, the nozzle 245, the nozzle 280 (and/or the nozzle 880), the power button 246, the one or more electrical indicators 247, and/or the electrical port 248 can be positioned for easy access (as an example, facing door 1202) when the material container 201 is locked to the retention assembly 230 inside of the storage container 1200. Other positions and/or orientations are contemplated for the actuator 231, the nozzle 245, the nozzle 280 (and/or the nozzle 880), the power button 246, the one or more electrical indicators 247, and/or the electrical port 248.

[0067] Benefits of the present disclosure include but are not limited to ease and simplicity of mechanically locking and automatically mechanically (e.g., magnetically) unlocking of material containers to and from retention assemblies; smaller and/or lighter material containers for easy and quick refilling; and reliable retention of storage containers to reduce or eliminate movement of the storage containers.

[0068] It is contemplated that one or more aspects disclosed herein may be combined. As an example, one or more aspects, features, components, operations and/or properties of the vehicle 100, the material container 201, the retention assembly 230, the nozzle 280, nozzle 880, the method shown in FIGS. 10A-10D, the retention assembly 230 implementation shown in FIG. 11, the vehicle panel 1100, and/or the storage container 1200 may be combined. Moreover, it is contemplated that one or more aspects disclosed herein may include some or all of the aforementioned benefits.

[0069] While the foregoing is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

What is claimed is:

1. A retention assembly, comprising:
 - an actuator mounted to a retention housing;
 - a locking component coupled to the actuator, the actuator movable to move the locking component between a locked position and an unlocked position; and
 - a magnet coupled to the actuator.
2. The retention assembly of claim 1, further comprising a bias element operable to bias the actuator toward the locked position, wherein the magnet is configured resist the bias element to hold the actuator in the unlocked position when an accessory is positioned on the retention assembly.
3. The retention assembly of claim 1, further comprising:
 - a rotatable link pivotably coupled to the locking component; and
 - a transfer link coupled between the rotatable link and the actuator.

4. The retention assembly of claim 3, wherein the rotatable link is rotatable relative to the retention housing, and the transfer link is pivotable relative to the rotatable link and the actuator.

5. The retention assembly of claim 3, wherein the rotatable link comprises an extension extending into an opening of the transfer link.

6. The retention assembly of claim 3, wherein:
the locking component interfaces with a first side of the rotatable link;

the retention assembly further comprises a second locking component interfacing with a second side of the rotatable link, the second side opposing the first side; and the rotatable link comprises an extension extending into an opening of the locking component.

7. The retention assembly of claim 3, wherein the actuator comprises:

a handle;
a receptacle having an opening, wherein the magnet is disposed in the opening; and
a flange abutting against the transfer link.

8. The retention assembly of claim 1, wherein the retention assembly comprises:

an extension comprising an outer surface and an opening formed in the outer surface;
a second actuator mounted to the retention housing; and
a second locking component disposed in the opening and coupled to the second actuator.

9. The retention assembly of claim 8, wherein the second actuator is movable to move the second locking component between a first position in the opening and a second position, wherein in the second position the second locking component is at least partially outside of the opening and abutting against a vehicle panel.

10. A material storage system, comprising:
a material container comprising a storage volume defined at least partially by a storage housing; and
the retention assembly of claim 1, wherein the material container is positionable to interface with the retention assembly, and the locking component moves between the locked position and the unlocked position to lock and unlock the material container to and from the retention assembly.

11. The material storage system of claim 10, wherein:
the material container further comprises:

one or more second magnets,
an extension extending relative to the storage housing, and
an opening formed in the extension, wherein the actuator is movable to move the locking component into and out of the opening; and

the retention assembly further comprises:

a bias element operable to bias the locking component toward the opening.

12. The material storage system of claim 10, wherein the retention assembly further comprises:

a nozzle operable to flow a material;
a pump fluidly connected to the nozzle;
a controller in communication with the pump; and
a battery electrically connected to the controller.

13. A material container, comprising:

a storage volume defined at least partially by a storage housing; and

a nozzle pivotably coupled to the storage housing, the nozzle comprising:

a flow housing at least partially defining an outlet opening, and

a shaft coupled to the flow housing and extending into a section of the storage housing, the shaft comprising a flow opening between the storage volume and the outlet opening.

14. The material container of claim 13, further comprising a washer seal disposed about the shaft of the nozzle, wherein the shaft is rotatable relative to the storage housing.

15. The material container of claim 13, wherein the nozzle further comprises a valve coupled to the flow housing.

16. A method of using a material storage system, comprising:

moving an actuator of a retention assembly to move a locking component of the retention assembly to an unlocked position to unlock a material container retained by the retention assembly, the moving of the actuator at least partially aligning a magnet coupled to the actuator with a second magnet coupled to the material container;

lifting a material container off of the retention assembly; moving the material container away from a vehicle; and dispensing a material from the material container.

17. The method of claim 16, wherein the retention assembly is coupled to a vehicle panel of the vehicle.

18. The method of claim 16, wherein the lifting of the material container moves the second magnet away from the magnet to bias the actuator to a locked position, and the method further comprises:

at least partially filling a storage volume of the material container with the material.

19. The method of claim 18, further comprising lowering the material container onto the retention assembly to lock the material container to the retention assembly.

20. The method of claim 16, further comprising connecting a spray nozzle to a nozzle of the retention assembly.

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